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Evaluation of Rule Acquisition for a New Speech Translation Method with Waveforms Using Inductive Learning

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ABSTRACT Conventional speech translation is an integration of three processing techniques: speech recognition, machine translation, and speech synthesis. Every part consists of several complicated parts. However, any technique has not obtained the satisfying results yet. Even if the accuracy of each processing achieves 90%, the final result is only 53% from the total results of speech recognition, morphological analysis, syntactic analysis, syntactic transform, speech synthesis, and more. Progress of each process is an only way to achieve total high accuracy because lack of accuracy in each process causes declining of the final result. In this paper, we propose a new speech translation method that is not dependent on individual character of any language. The structure of proposed translation system is much simpler than others due to omitting several complicated processes to realize speech translation. We focused on the time-varying characteristics of acoustic speech waveform of source and target languages, and our method realize the speech translation by learning translation rules that have correspondence between two utterances inductively. The approach will be able to not only translate speech samples with independence of any language, but also will deal with all processes in short time because the system does not have a lot of processing stages. In this paper, we deal with a machine translation between Japanese and English.

KEY WORDS
Speech translation, Inductive learning, Time-varying characteristics of speech

1. Introduction

In conventional approaches, speech translation consists of several complicated processing stages. Speech recognition part transforms speech to character strings: phoneme, syllables or words. In machine translation part, a result is conducted by morphological analysis, syntactic analysis, and parsing of the sentence of the target language. And finally, speech synthesis part outputs translated speech. Figure1 shows a whole procedure of conventional speech translation approach.

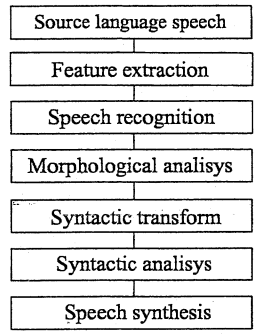


Figure 1. Conventional approach to speech translation processing

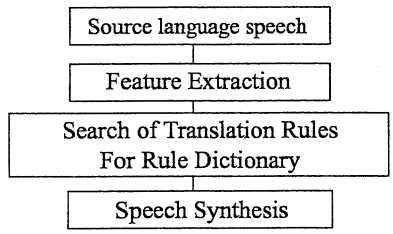


Figure 2. Procedure of our approach.

However, our approach has much simpler structure than others[1] because we thought that simplification has an ability to prevent decreasing of the translation precision. In our past studies, we attempted to recognize several vowels and consonants using Neural Networks that had different structures with TDNN[2], however we could not obtain high accuracy of recognition[3]. Then, we focused on speech waveform itself, not character strings obtained by speech recognition. We decided on dealing with correspondence of time-varying characteristics of speech waveform instead of characters between two utterances.

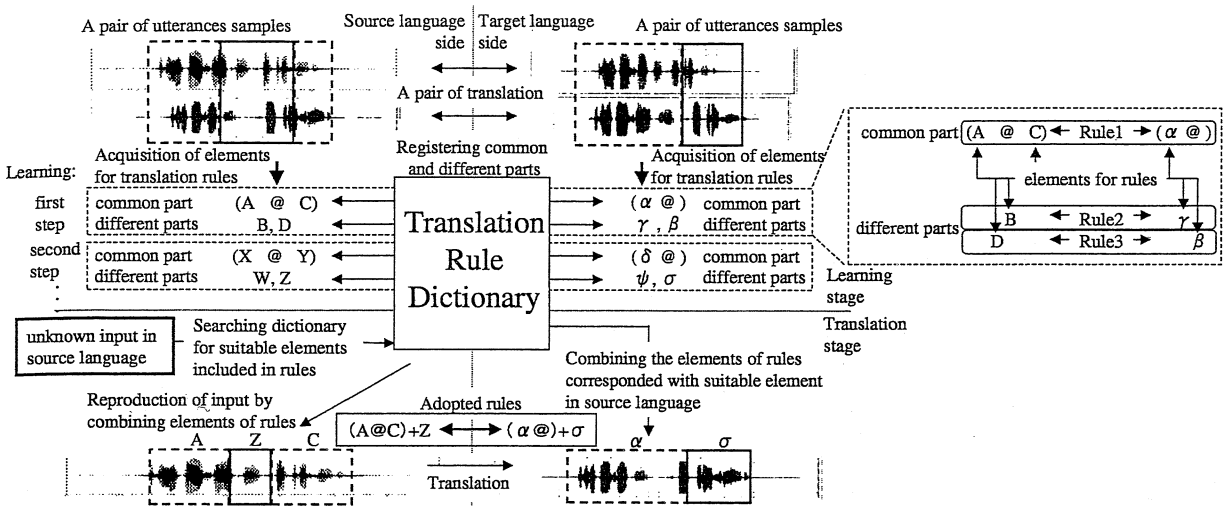


Figure 3. processing structure.

In conventional approaches, it is indispensable that machine translation part must be changed if target language is changed. Because morphological analysis and syntactic analysis in this part depend on each individual character of languages strongly. However, in our approach, the system can be applied for all languages, because there is no stage depended on any specific language. Any difference on language is not able to affect on the ability of system, fundamentally because we focus on the time-varying characteristics of speech, not on the original character of languages. It is very important that we realize speech translation with new approach that is independent of individual character of any language.

Figure2 shows the processing stages of our method and Figure3 shows the overview of translation procedure of proposed system. The correspondence of meaning between two languages is being decided by human. We compare two examples of time-varying characteristics of speech between two utterances of the same language, then we extract common and different parts. The translation rules are created and registered in the translation rule dictionary. The elements of translation rules keep location information of corresponded common and different parts in utterances of each language. They are acquired not only by comparing speech samples but also by comparing them using the inductive learning[4]. In a translation phase of unknown utterance of source language, the system checks up common and different parts within the unknown utterance comparing it with speech waveforms of source language, stored as examples in the system. A speech waveforms of target language will be selected according to translation rules.

The boundaries of word, syllable, or phoneme are not important for our method because we do not transform the acoustic characteristics of speech to character strings like other speech recognition methods. The special feature of our approach is utilization of the correspondence of acoustic common and different parts in speech samples. Segmentation to phonemes, syllables and so on, is unnecessary in our system.

We focus on the time-varying characteristics of speech. We extract common and different parts by comparing them for determining rule elements of rules. At the time, a transformation of time-varying characteristics of speech is performed by translation rules. The rules are registered in the rule dictionary. The inductive learning keeps translation rules that have common or different parts acquired by comparing characteristics of speech. Furthermore, the inductive learning also obtains translation rules recursively by comparing the rules registered in the dictionary even with a small amount of samples. This is one of the merits of the inductive learning. The repetition of this learning process is able to increase the number of rules what makes the system more flexible for translation process.

2. Speech processing

2.1 Speech data

We extract time-varying spectral characteristics in speech and utilize them for experiments. We recorded speeches with 48kHz sampling rate on DAT, and downsampled them to 12kHz. All speech data is spoken by Japanese male students of our laboratory. We obtained 60 sentences read 2 times, and speech data were divided into 2 groups of 30 sentences,

In our approach, we utilize the vector of time-varying characteristics of speech. The conditions for speech analysis, are shown in Table1. And we adopt

Table 1. Experimental conditions of speech processing.

Size of frame	20msec
Frame cycle	3msec
Speech window	Hamming Window
AR Order	14

the same conditions and the same kind of characteristics parameters of speech throughout the experiments because we need to obtain acoustic characteristics that are not dependent on any language.

2.2 Spectral characteristics of speech

We need finding common and different parts by comparing spectral characteristics of speech in our approach. In this report, Spectral envelopes derived from FFT of LPC cepstrum parameters are used as spectral parameters, because we could obtain better results than other characteristics of speech: Spectral envelopes derived from LPC cepstrum coefficients, envelopes of Mel cepstrum from LPC analysis, and envelopes of Mel cepstrum from direct mel-cepstrum analysis.

2.3 Start point search of common part between utterances

This section describes how the start point of common and different parts are decided by comparing two utterances.

Generally, normalization of the elasticity on time domain is needed when speeches are being compared. We try to consider a method that is able to obtain a result similar to dynamic programming[5][6] to solve the difference as above.

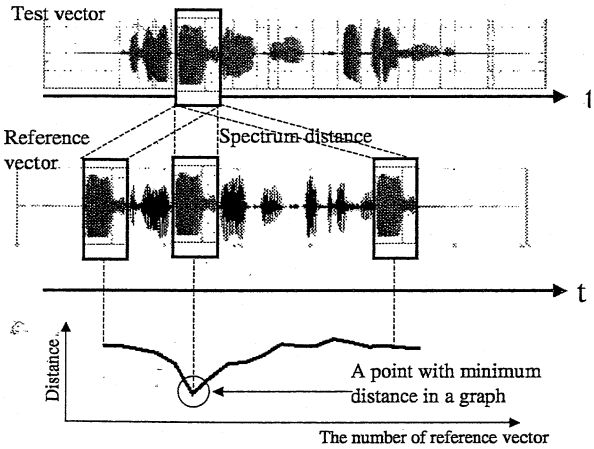


Figure 4. Comparison of vector sequences.

We adopted a method for calculate the distance between characteristic vectors of speeches. A sequence of characteristic vectors named "test vector" is picked up from test speech with a window that has fixed length. At the time, the other sequence of characteristics vectors, named "Reference Vector" is picked up from reference speech. A distance value is calculated comparing the current "test vector" and the current "reference vector". Then, the reference vector is shifted by a constant interval, and a new distance value is calculated again. Repeating these procedures, we get a sequence of distance values(shown in Figure4). If there is a portion that has similar characteristics vector as the "test vector" in the reference speech, the sequence of distance value should have a obvious valley-like local minimum point. Next, we shift the "test vector" in the test speech, repeat the calculation procedure mentioned above, then we get another sequence of distance value for new a "test vector".

Examples are shown in Figure5, 6, and7. In those figures, italic fonts express Japanese. Vertical axis shows value of distance, and horizontal axis shows the shift number of reference vector, i.e., the position of the reference vector. When the "test vector" locates at the top of the test speech, we get a curve drawn in most lower position. When the "test vector" at the end of the test speech, a curve drawn in most upper position is obtained.

As shown in Figure5, when the contents of the test and reference speech are the same, the obvious local minimum points appear sequentially in each distance curve. Meanwhile, as shown in Figure7, if the contents of test and reference speech are completely different, there is no obvious local minimum in any distance curve. Thus, if there is a position that obvious local minimum points appear sequentially, that portion can be considered as a "common part".

2.4 Decision of obviousness of the local minimum point

To judge the existence of "obviousness" of the local minimum point in the distance curve, we introduce statistical information. For single curve of distance, we calculate mean value, the smallest value, and variance of distance values within the curve. The threshold θ is calculated as $\theta = 4\sigma^2$ from the equation of Gaussian distribution. This threshold represents a range of spectrum simirality from the mean value of distance within a curve.

If $(x - m)^2 > \theta$, then we represent the common parts with "0", else we repesent as different parts with "1", where x shows the smallest distance point and m shows mean value of distance, respectively.

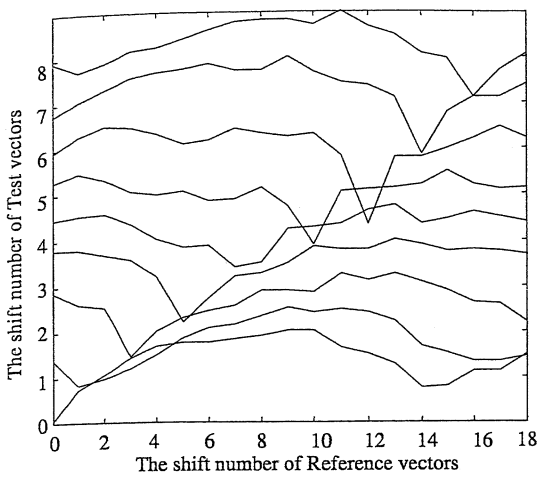


Figure 5. Difference between utterances (1). *"KOREWA WATASHINO OGORIDESU"* (This is my treat)

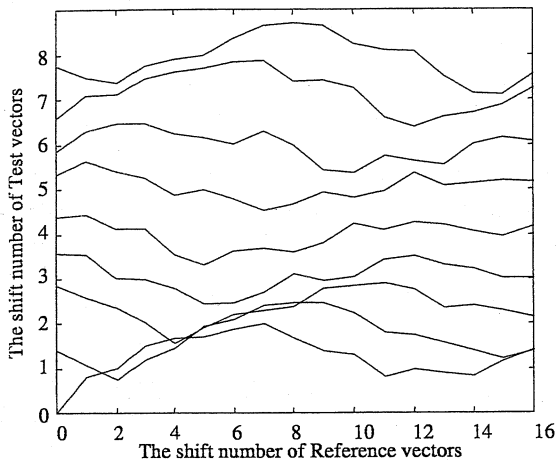


Figure 6. Difference between utterances (2). *"KOREWA WATASHINO OGORIDESU"* (This is my treat) - *"KOREWA ANATANI DESU"* (This is for you)

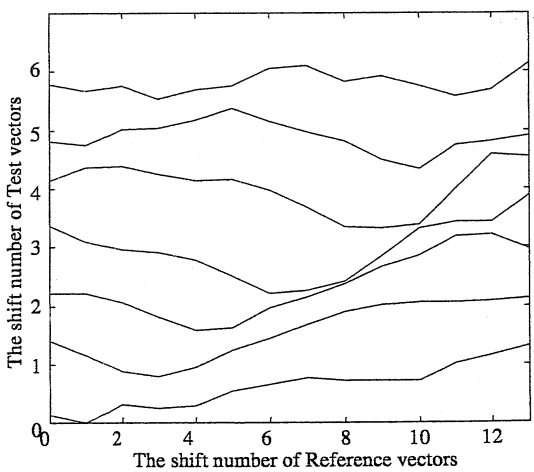


Figure 7. Difference between utterances (3). *"OKANJOWO ONEGAI SHIMASU"* (Check, please) - *"CHOSHOKUWA NANJI DESUKA"* (What time is breakfast)

3. Extraction of translation rules and inductive learning

3.1 Extraction of common part for translation rules

Start points calculated in above section are applied for determining rule elements needed to create translation rules. We prepare speech characteristics for each language to decide the correspondence between common and different parts because each kind of parts contains different information about elements for translation rules. Three types of compared examples are shown in Figure8, 9, and 10. These types represent the compared sentences structure and are registered in a translation rule A continuous values of "0" represents a common part and a continuous values of "1" represents a different part, respectively as in Figure9. A whole utterance is calculated as a common part in Figure8.

In Figure9, the parts corresponded with *"KOREWA"*(This) and *"DESU"*(is) as we assumed are calculated as common parts, and the parts corresponded with *"AKAI"*(red) and *"ANATANI"*(for you) are treated as different parts. At that point, correct boundaries of common and different parts in time domain are lost as the resolution of frame rate for each vector. However, it is not important for us because we are aiming to obtain translation that meaning can be understood.

In the Figure10, the whole part is calculated as a difference in contrast to Figure8. These comparison results are comparable with lexical contents because the lexical sentence structure is represented as "common-different-common". In all above mentioned cases, we acquire sets of common and different parts if those utterances were perfectly matching or did not matched at all. The translation rules are being obtained by combining sets of common and different parts of both source and target utterances.

At the last stage, these translation rules have elements as bellow. They are acquired if the types of sentences from each language are correspondent. Obtaining translation rules is impossible when the types of sentence structures are different because the correspondence relation between two languages samples can not be decided. Acquired rule types are as follows:

1. those with very high sentence similarity
2. those with sentences consisting common and different parts
3. those with very low sentence similarity

Information acquired from such rules is as follows:

- above mentioned type of translation rule

Shift number	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Result	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Figure 8. Common and different parts (1). *"KOREWA ANATANI DESU"* (This is for you)

Shift number	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Result	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0

Figure 9. Common and different parts (2). *"KOREWA ANATANI DESU"* (This is for you) - *KOREWA AKAI DESU"* This is red

Shift number	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Result	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Figure 10. Common and different parts (3). *"KOREWA ANATANI DESU"* (This is for you) - *WATASHIWA YAKYUWO SURU"* I play baseball"

Table 2. Conditions for experiments.

Frame length of test vector	400msec
Frame rate of both vectors	50msec
The number of open data	10 sentenes
The rate of agreement for adopting rules	95%

Table 3. Translation rules.

	extracted rules	registed rules
set1	3363	2908
final	13725	12196

- index number of an utterance
- sets of start and end points of each common and different part
- index number of a target language's utterance sample

3.2 Inductive learning

The inductive learning that we proposed obtains rules by extracting common and different parts through the comparison between two samples. Moreover, this method can also acquire rules by repetition of learning in registered translation rules, recursively. If some rules that are not utilized in translation stage many times, they will be eliminated for generalizing the rule dictionary optically. If new common and different parts are found during comparing the rules, they will become elements of new rules. The new rules have shorter parts than previous ones, and we assume that they are much more reliable for translation.

This recursive acquisition of rules is implemented for rules containing not only common parts, but also different parts because we assume that there is a possibility of information for translation included also in different parts. It can be an advantage of less examples that have to be prepared beforehand because in conventional approaches based on examples, a lot of sample data have been needed to acquire a large amount of translation rules and to accomplish the speech translation with high accuracy.

4. Evaluation experiments

4.1 Experiment of searching for start points and acquisition of translation rules

All data used in experiments is obtained through several speech processes described in 2.1. Table2 shows the conditions for these experiments. We prepared 2 groups of speech samples, and each set is consists of 30 sentences. The utterances in set1 *"KOREWA * DESU"* mean (This is *), and those in the other set *"SOKONI * GA ARIMASU"* have the meaning of (There is *). The parameters on frame are determined by results of some preliminary tests.

In these experiments, we extracted many sets of common and different parts by comparing time-varying characteristics of speech in each language. Translation rules are obtained by using sets of start and end points. Table3 shows the number of acquired and registered rules. The registered rules have correspondence between two languages.

Table 4. Adopted rules.

Sentence	Rule Type	Corresponded Part	Speech
S2-12J	common@common	(0-110)@(210-300)	KOREWA ZASSHI DESU
S2-11J	commuon@common	(0-110)@(210-300)	KOREWA SHORUI DESU
S1-18J	difference	(120-210)	SOKONI KAGIGA ARIMASU
S1-17J	difference	(120-210)	SOKONI KUNIGA ARIMASU
S2-12E	common@	(0-180)@	This is a magazine
S2-11E	common@	(0-180)@	This is a document
S1-18E	difference	(150-200)	There is a key
S1-17E	difference	(150-200)	There is a tree

4.2 Result for experiment of translation and discussion

If the whole input sentence can be replaced at source language side with elements already existing in dictionary, the sentence will be able to be translated without losing the meaning because each replaced part has correspondence with its' equivalent in the target language. However, not all elements can be used for every translation so the system determine the most appropriate candidates. If the level of similarity is lower than a rate of agreement as Table2, the element can not become a candidate for actual translation.

We succeeded to obtain proper translation results during our experiment. Table4 shows the applied rules for translation in this experiment. We inputted an open data which was Japanese sentence *"KOREWA KAGI DESU"*(This is a key). First, the system calculated some suitable rules that had high similarity with common parts by comparing a whole input sentence with the elements of all rules. Afterward, the system calculated suitable rules for not yet replaced segments by comparing input sentence with elements that are registered in the dictionary as different parts. Finally, the translation result "This is a key" was combined with suitable elements of rules that previously became candidates.

In this result, common and different parts were calculated in rough by comparing sample utterances even though the accurate boundaries of words, syllables or phonemes had not be utilized. This successful result certified that the system was able to work regardless of individual character for specified languages.

We need to obtain more appropriate translation outputs by decreasing the number of candidates rules. A large number of candidates proposed by the system causes much trouble for a user, so we consider adopting some algorithm for solving this problem in near future. Furthermore, we must apply a new method that will let us decrease the calculation cost because the learning process is very time-consuming.

5. Conclusion

In this paper, we acquired acoustic common and different parts of utterances, translation rules obtained from them, and the results of whole translation through the proposed approach. The new method is independent of any individual character of given languages. Furthermore, we will consider an algorithm for selecting correct candidates of translation rules, we need also to implement the speech synthesis section to confirm the ability of our system, and finally we will carry out experiments with a larger amount of spoken data for achieving satisfying number of translation rules.

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